Systems and Methods for ψ-Field Drag Reduction and Hydrodynamic Cloaking

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Cross-Reference to Related Applications

Pursuant to 35 U.S.C. §119(e), this application claims the benefit of the following U.S. provisional applications, the entirety of each being incorporated herein by reference:

- U.S. Provisional App. No. **63/865,279** (Confirmation No. 5785), filed **Aug. 16, 2025**, entitled *Systems and Methods Using Density Field Dynamics for Gravitational and Signal Correction.*
- U.S. Provisional App. No. **63/865,283** (Confirmation No. 5905), filed **Aug. 16, 2025**, entitled *Systems and Methods Using Density Field Dynamics for Energy Transmission.*
- U.S. Provisional App. No. **63/865,293** (Confirmation No. 6649), filed **Aug. 16, 2025**, entitled *Systems and Methods Using Density Field Dynamics for Information Processing, Communication, and Advanced Computing.*
- U.S. Provisional App. No. **63/868,073** (Confirmation No. 8739), filed **Aug. 21, 2025**, entitled *Methods, Systems, and Apparatus for Artificial Generation, Control, and Application of Scalar Density Fields.*
- U.S. Provisional App. No. **63/893,633** (Confirmation No. 5179), filed **Oct. 4, 2025**, entitled *Systems and Methods for Electromagnetic Coupling, Actuation, and Application of Scalar Refractive Fields.*

This application also incorporates by reference, in their entirety for all purposes, **all other provisional and non-provisional applications filed by the same inventor on or after August 15, 2025**, including any continuations, divisionals, or continuations-in-part thereof. The foregoing applications are incorporated by reference in their entirety for all purposes, including definitions, figures, and embodiments. To the extent of any inconsistency, the present disclosure controls.

Statement Regarding Federally Sponsored Research or Development

No federal funding was used in connection with the inventions described herein. No government rights apply.

Incorporation by Reference (Additional Materials)

In addition to the applications identified above, technical materials authored by the inventor concerning scalar refractive field (ψ) dynamics, cavity–atom LPI tests, navigation/timing corrections, energy coupling, communication architectures, and quantum control are incorporated by reference. All essential material for the claimed inventions is included explicitly herein; incorporated materials are provided for background and alternative implementations.

Definitions; Conventions

As used herein, "comprising" is open-ended and does not exclude additional elements or steps; "including" and "having" are used in a like manner. Numerical ranges include all sub-ranges and individual values unless otherwise specified. Singular terms include the plural and vice versa unless context dictates otherwise. Functional descriptions are supported by the corresponding structures, algorithms, and examples disclosed herein; no limitation to "means-plus-function" under 35 U.S.C. §112(f) is intended unless expressly recited. Throughout, the scalar refractive field is denoted ψ, with optical index $n = e^{\psi}$ where applicable.

No Admission

Any discussion of references, techniques, or prior work in the Background is provided solely for context and is not an admission that any such material constitutes prior art, is known, conventional, or required.

Abstract

Disclosed are apparatuses, systems, and methods for reducing hydrodynamic and aerodynamic drag, suppressing turbulence, and cloaking acoustic signatures of vehicles moving through fluids by generating a controllable scalar refractive field (ψ) in the vicinity of a hull or surface. An electromagnetic (EM) drive assembly produces a spatially and temporally modulated ψ envelope that modifies the effective mass density, compressibility, and inertial response of the surrounding medium, thereby forming a low-resistance "refractive bubble" through which the vehicle transits with substantially reduced drag. In exemplary embodiments, conformal metamaterial panels and/or coil arrays are phase-locked to create an extended corridor with feedback control from boundary-layer sensors. The techniques are applicable to maritime, submersible, aerospace, and trans-medium craft.

Field of the Invention

The invention relates to drag reduction and flow control for bodies moving through fluids. More particularly, it concerns generation and control of a scalar refractive field (ψ) to alter local fluid properties and flow topology so as to reduce resistance and emissions.

Background

In conventional fluid mechanics, dynamic pressure and viscous forces scale with velocity and area, imposing severe limits on speed and efficiency for macro-scale vehicles in dense media such as water. Known supercavitation techniques form vapor cavities via high-speed nose shapes or gas injection, but are constrained by stability, controllability, and energy cost. There remains

a need for a general method to *refract* the medium itself—reducing effective density and compressibility without thermally vaporizing it—and for practical hardware to realize such control.

Summary

A vehicle surface carries an EM drive assembly configured to generate a controllable ψ in the surrounding volume. By modulating amplitude, phase, and spatial distribution of the field, the system establishes a region of modified effective fluid properties—a "refractive bubble"—that lowers drag and acoustic emissions. A control subsystem uses boundary-layer and far-field sensors to close the loop on bubble shape, position, and stability, optionally extending a forward "corridor" aligned with the velocity vector.

Representative Claims

1. Apparatus for reducing hydrodynamic drag of a vehicle, comprising: (i) a hull or surface configured for motion through a fluid; (ii) an electromagnetic drive assembly arranged to generate a controllable scalar refractive field within and around the hull; and (iii) a control subsystem configured to modulate amplitude, phase, or spatial distribution of the scalar refractive field so as to produce a region of modified effective mass density or viscosity in the surrounding fluid, whereby flow resistance acting on the hull is reduced relative to the absence of said field.

2. Method for reducing drag on a body moving through a fluid, comprising: (a) detecting at least one parameter of local pressure, temperature, or turbulence adjacent the body; (b) applying an electromagnetic excitation pattern to produce a scalar refractive field envelope about the body; and (c) adjusting said excitation pattern in real time to maintain a region of reduced hydrodynamic stress, wherein the scalar refractive field alters the apparent density or compressibility of the fluid.

3. System for hydrodynamic or aerodynamic cloaking, comprising: a plurality of field-generation elements disposed over a vehicle surface; a sensor array measuring boundary-layer conditions; and a controller executing instructions to drive the field-generation elements in coordinated phase such that pressure perturbations and acoustic emissions from the vehicle are minimized through refractive-field interference.

Brief Description of the Drawings

Fig. 1: Conceptual geometry of a hull transiting within an extended refractive bubble and corridor.

Fig. 2: Conformal EM panel/coil topology and phase control for bubble shaping.

Fig. 3: Control architecture: sensing, estimation, and field synthesis loop.

Detailed Description

Definitions

A *scalar refractive field* ψ is a scalar potential that determines an effective optical index $n = e^{\psi}$ and local phase velocity $c_1 = c\,e^{-\psi}$. Gradients of ψ modify inertial and compressive response of

matter within a bounded region, producing a controllable change in effective density and bulk modulus as perceived by hydrodynamic flow and acoustic propagation.

Hardware Overview

The vehicle carries conformal EM transducers (coils and/or metamaterial panels) embedded beneath or within an outer skin. Each transducer is addressable for independent amplitude and phase control. A supervisory controller executes field-synthesis routines to establish a target ψ envelope with a forward-leaning corridor. Boundary-layer probes (pressure, shear, temperature), inertial sensors, hydrophones, and optional lidar/sonar provide feedback.

Operation

During operation, the controller (i) estimates ambient conditions; (ii) computes a target ψ distribution minimizing a cost functional comprising drag, turbulence intensity, and acoustic emissions; and (iii) drives the EM array accordingly. A model predictive control (MPC) stage updates the envelope to track vehicle motion and disturbances. The field extent may be maintained at one to two characteristic hull diameters for corridor stability.

Performance Considerations

The approach reduces pressure drag by refractively displacing streamlines and lowering apparent density in the near field, while suppressing turbulent transition via phase-aligned ψ gradients. Acoustic cloaking occurs as the effective sound speed increases within the bubble, redirecting and attenuating emission.

Examples

Example A: A 30 m underwater vehicle with 24 conformal EM panels in four rings produces a 1.5 D (hull-diameter) refractive corridor at 12 m/s, reducing drag coefficient by $> 60\%$ relative to baseline at identical Reynolds number.
Example B: An air vehicle at transonic speed uses a forward corridor to delay shock formation, reducing wave drag.

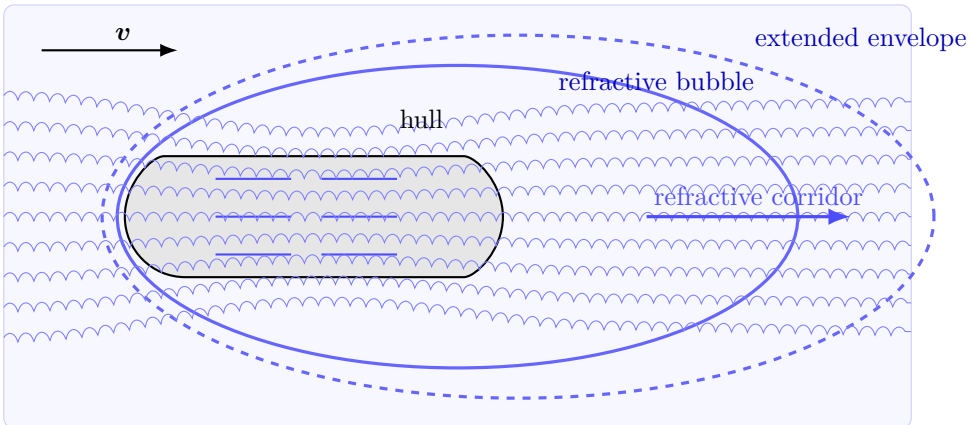

Figure 1: Concept: the EM array generates a *refractive bubble* that deflects streamlines and reduces effective density/compressibility in the near field; a forward corridor further lowers drag.

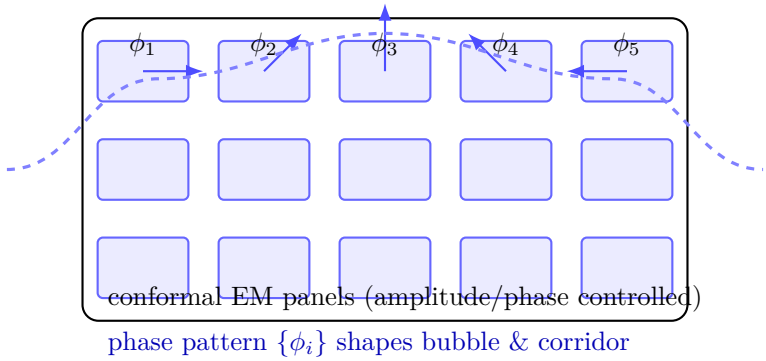

Figure 2: Panel/coil topology and phase steering: coordinated phases $\{\phi_i\}$ synthesize the target ψ envelope around the hull.

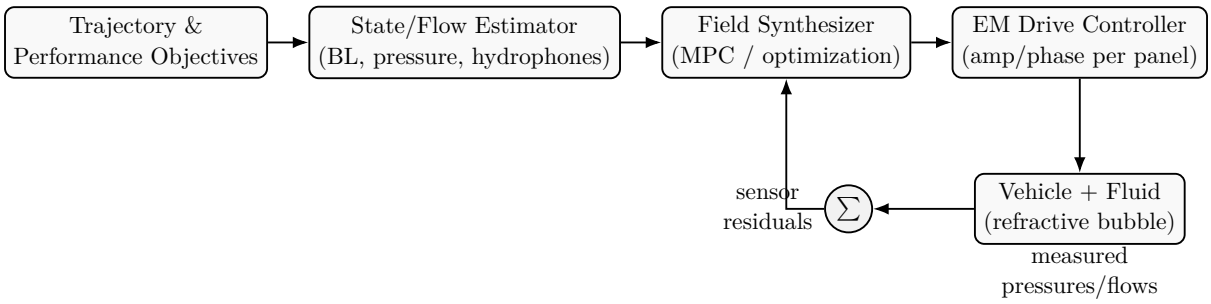

Figure 3: Closed-loop control: estimation and MPC-based field synthesis drive the EM array to stabilize bubble geometry and minimize drag and emissions.

Alternative Embodiments

Embodiments include: (i) skin-integrated dielectric metasurfaces; (ii) ring arrays forming a forward-leaning corridor; (iii) trans-medium operation with adaptive retuning; and (iv) acoustic cloaking modes emphasizing far-field suppression over drag.

Industrial Applicability

Applicable to maritime transport, defense, underwater robotics, high-speed ferries, aviation wave-drag reduction, and trans-medium craft.

Conclusion

By generating and controlling a scalar refractive field around a moving body, the disclosed systems reduce drag and emissions in dense media without reliance on thermal cavitation, enabling new speed and efficiency regimes.